Technical Document #246: Database Systems - Comprehensive Overview

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Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Relational databases organize data into tables with rows and columns.
- Database connection pooling reuses database connections to reduce overhead.
- Sharding distributes data across multiple database servers.